Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/26/2006
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Project Title:	The Genetic Basis of the Loss of Musculo-Skeletal Tissue during Weightlessness: Towards the Identification of Individuals that are at Greatest Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	01/27/2003	**End Date:**	01/26/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 1/26/2007 per J. Dardano (5/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Leah Rae (The Jackson Laboratory) Rubin, Clinton (SUNY Stony Brook)		
Grant/Contract No.:	NAG9-01499		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The National Research Council's Space Studies Board has stated that a principal physiologic hurdle to man's extended presence in space is the osteopenia and sarcopenia which parallels reduced gravity. The extent of the loss is extremely high, approaching a decrease in bone mineral density (BMD) in the lower appendicular skeleton at a rate of 1.6% per month and reducing maximal voluntary contractions of some muscle groups at a rate of 5% per month. Interestingly, the amount of bone (and muscle) loss between individual astronauts is highly variable with some astronauts losing large amounts of tissue while others are largely unaffected. This large individual variability, which has also been observed during bedrest and immobilization studies on Earth, may be accounted for, at least to a large extent, by genetic variations which may give rise to a differential mechanosensitivity of the musculoskeleton. We have collected preliminary data demonstrating that genetically distinct inbred strains of mice also demonstrate a distinct sensitivity to conditions of simulated weightlessness; while as much as 60% of trabecular bone is lost in the hindlimbs of BALB/cByJ mice within 3 weeks of disuse, the same conditions leave trabecular bone quantity and quality nearly unchanged in C3H/HeJ mice. In this proposal, we aim to elucidate this genetic basis of the sensitivity of the musculo-skeleton to the loss of appropriate mechanical signals by identifying the quantitative trait loci (QTL) responsible for the difference exhibited between these two strains of inbred mice. The identification of QTL (and ultimately the responsible genes) may be used as both a critical diagnostic sensor for the identification of astronauts that are in greatest need of pharmacologic and/or biomechanical countermeasures in space and as a discovery tool of novel drug targets for the preventing the loss of musculo-skeletal tissue. - to reveal the chromosomal locations (genes) that are associated with a high and low sensitivity of bone and muscle to mechanical disuse (tail suspension) by crossing a mechanically responsive (BALB/cByJ) and a mechanically unresponsive strain (C3H/HeJ), thus determing at the genetic level why only one of the two mouse strains responds to the loss of physical factors
- to study whether the tight association between muscle and bone will cause QTLs for muscle to overlap with QTLs for bone.

- to develop a genetic diagnostic sensor to identify individuals who at highest risk of loosing bone and muscle due to space flight, bedrest, or aging.

- to investigate three-dimensional changes in trabecular bone architecture induced by mechanical disuse which may provide the basis for future finite element (FE) models to study the mechanical mechanisms of bone loss.

- to probe the influence of genetics on the recovery of bone loss upon a period of unloading.

- to serve as a discovery tool for novel drug targets that will prevent the loss of musculo-skeletal tissue. The very large variability in bone loss between astronauts (i.e., some lose a lot, some lose almost nothing), in the absence of large variations in environmental factors (e.g., diet or exercise), suggest a strong genetic influence on bone’s susceptibility to the loss of functional weightbearing. The principal aim of this proposal to elucidate the genetic basis of the sensitivity of the musculo-skeleton to the loss of appropriate mechanical signals may lead to the design of a diagnostic sensor for the identification of astronauts that are in greatest need of pharmacologic and/or biomechanical countermeasures in space, as well as to the discovery of novel drug targets that will allow for safe and efficacious countermeasures. This proposal addresses two questions that are central to the Critical Path:

Question 1g: What are the important predictors for estimating site-specific bone loss and fracture risk during hypogravity exposure, including, but not limited to ethnicity, gender, genetics, age, baseline bone density and geometry, nutritional status, fitness level and prior microgravity exposure? The focus of this grant is to investigate how genetics modulates bone quantity and micro-architecture and, more specifically, to identify the specific chromosomal regions that predispose individuals to the loss of muskulo-skeletal tissue in space. Knowledge of these chromosomal regions (QTL) may be used as a diagnostic tool before space travel (and the loss of tissue) and for the design of individually tailored treatment regimes. Further, as described below and in last year’s report, we are also investigating how gender, baseline density and geometry affect site-specific bone loss.

Question 1k: Are hypogravity-induced changes in bone density, geometry, and architecture reversible upon encountering partial gravity exposure, or on return to full gravity (1-G)? Mice that are subjected to hindlimb suspension in this study are allowed, upon completion of 3wk of unloading, to resume normal cage activities and any restoration in bone morphology and architecture is assessed by in vivo micro-computed tomography.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Skeletal pathologies involving bone loss, such as osteoporosis, are having a catastrophic impact upon the quality of life of the affected population and upon the cost of North American health care. In the United States alone, it is estimated that more than 44 million women and men (that is 55% of all people older than fifty years) are inflicted with osteoporosis or will be at high risk of developing it, currently costing the health care system more than $20 billion annually. Data derived from these studies will not only be relevant to spaceflight but be equally important to the prevention and treatment of bone loss associated with bedrest, musculo-skeletal diseases, and decreased levels of physical activity in our aging population.

Progress over the Past 12 Months
Over the past year, we have finished the experimental part of this grant which involved the suspension of 500 female F2 mice. Currently, we are analyzing genomic DNA extracted from these genetically distinct mice and relate differences in genetic markers to differences in disuse-induced changes in bone and muscle morphology to determine the quantitative trait loci (QTL) that regulate the susceptibility of the musculo-skeleton to unloading and re-ambulation. We expect to complete this work by January 2007. Below, we present an update on the phenotypic relation between unloading and re-ambulation.

Introduction

Disuse induced osteopenia and the associated loss of bone strength present a major socio-economical burden and can lead to major complications both in space as well as on earth. The deterioration of bone mass and morphology in astronauts, cosmonauts, and bedrest patients is marked by large variations between individuals that may implicate a strong influence of genetic make-up on the magnitude of the response. This hypothesis is corroborated by well-controlled mouse studies in which environmental and genetic variables can be carefully controlled. Among genetically homogeneous inbred strains of mice, for instance, BALB/cByJ mice, respond to mechanical unloading with a

Task Progress:	60% reduction in trabecular bone volume fraction while C3H/HeJ mice are comparatively unresponsive. Upon completion of bedrest or upon return from space to earth, humans only partially recover the lost bone mass and may initially even continue to lose bone. During this recovery period, similar to the unloading phase, relative changes in bone mass are strongly dependent on the individual, perhaps arising from genetic variations between subjects. Whether those individuals that suffer from the (genetically determined) greatest (or smallest) losses are also the ones who are most (or least) successful in recovering tissue during reambulation is unknown. If such a relationship exits it would provide a tool for the selection of the most appropriate recovery treatment and may also offer critical information on the genetic regulation of bone recovery (e.g., are the genes associated with a high susceptibility to disuse induced bone loss identical to the genes that define the addition of bone mass during recovery?) Here, using mice from a genetically heterogeneous population, the hypothesis that the genetically determined rates of bone recovery upon disuse are linked to the rates of bone loss during unloading was tested. Methods Forty-five female, second generation (F2) BALBxC3H mice (4mo old) were subjected to 3wk of disuse (hindlimb suspension) followed by 3wk of reambulation. All mice were genetically distinct from each other and resulted from crossing a mouse strain with a high skeletal mechanosensitivity (BALB) with one whose skeleton displays a low sensitivity to changes in its mechanical environment (C3H). In-vivo micro-computed tomography (17 um resolution) quantified longitudinal changes in trabecular bone morphology of the left and right distal femur at baseline, after 3wk of unloading, and after completion of the unloading/reambulation protocol. At these scan settings, we have previously shown that ionizing radiation produced by the scanner does not interfere with changes in bone morphology induced by disuse. Bone volume fraction (BV/TV), bone surface (BS), trabecular separation (Tb.Sp), trabecular thickness (Tb.Th), trabecular number (Tb.N), connectivity density (Conn.D), and the structural model index (SMI) were determined for all three time points in each mouse and averaged across the left and right leg. Repeated measure ANOVA tested for differences in bone quantity and architecture between the three time points. Linear regression analyses correlated baseline morphology and morphological changes during the disuse period to changes in morphology during the recovery period. All experimental procedures were approved by Stony Brook's Institutional Animal Care and Use Committee. Results Over the 3wk disuse period, bone volume fraction (BV/TV) in the distal femur diminished on average by 37%±16% ($p<0.0001$). Supporting the genetic influence on this large variability in bone loss, data were normally distributed with losses ranging from 4.5% to 72%. Trabecular bone architecture also deteriorated dramatically with unloading; connectivity density decreased by 33% ($p<0.0001$), trabecular number by 13% ($p<0.0001$), and trabecular thickness by 17% ($p<0.0001$). Changes in bone volume fraction correlated with changes in trabecular number ($R2=0.46$), connectivity density ($R2=0.52$) and trabecular thickness ($R2=0.79$). The large variations in genetically determined baseline bone volume fraction showed a significant, but inverse, relation with the magnitude of bone loss ($R2=0.21$). The baseline ratio of bone surface to bone volume (BS/BV) had a low association with the relative amount of tissue loss ($R2=0.13$), similarly to the non-significant relation between the amount of baseline BS and the absolute loss in BV/TV ($R2=0. 09$). Changes in trabecular bone volume fraction during the 3wk reambulation period ranged from a continuation of bone loss, to cessation of catabolic activity, to a partial reversal of the previous bone loss. Changes in BV/TV during this period (with respect to the start of reambulation period) ranged from –18% to +73%. Across all mice, bone volume fraction and trabecular thickness increased by 10%±20% ($p<0.0001$) and 19±10% ($p<0.0001$) respectively while trabecular number (-11±10%, $p<0.0001$) and connectivity density (-22±32%, $p<0.0001$) continued to decrease. When separated into "losers" and "gainers", 14 out of 45 mice continued to shed bone at a rate of 3±2% per week while the other 69% re-gained 54±64% of their previously lost bone volume. For any given individual, the amount of bone lost during the disuse period or the amount of BV/TV at completion of disuse were poor predictors ($R2=0.02$ each) of the relative amount of bone gained during the reambulation period. Discussion Similar to data from astronauts and bedrest patients, changes in bone morphology during disuse and reambulation were associated with large variations in this genetically heterogeneous mouse population that resulted from double crossing two inbred strains of mice with very distinct skeletal responses to the loss of functional weightbearing, Even though most mice were able to recover part of the tissue mass that was lost during disuse, this recovery occurred by re-thickening individual trabeculae while the connectedness of the structure further deteriorated. The amount of tissue recovery was neither associated with any baseline bone morphology parameter nor with the rates of tissue loss induced by disuse. These data indicate the strong influence of genetic variations on the changes in bone morphology induced by rapid changes in its mechanical milieu, emphasize that is critical to preserve bone's micro-architecture before individual trabecular connections are lost, and suggest that the genes modulating changes in morphology during disuse are, at least in part, distinct from those regulating tissue recovery. The current identification of these sets of genes may allow the identification of those individuals that are most susceptible to bone loss and the least capable of recovering.
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